Survival chances in the RMS Titanic wreck

Valeria Angelino

Introduction

The sinking of the RMS Titanic in the Atlantic Ocean is one of the greatest maritime disasters in history, claiming over 1500 lives (Ocean Learning Hub, 2026). There have been conversations about the ship's shortage of emergency equipment, the chaotic execution of a "women and children first" protocol and overall inadequate evacuation procedures (Royal Museums Greenwich, 2026). However, were there any key factors influencing who survived? Would women and children have a higher chance of surviving? What about people in first-class compared to those in third-class? This paper aims to explore these questions using the *Titanic* dataset, which includes whether a passenger survived, their sex and age, their passenger class and cabin number, and whether they were travelling with a family member (child, sibling, parent). The course instructor provided the dataset via Canvas, and the analysis was done using RStudio.

Exploratory Data Analysis

Firstly, by looking at the dataset, I noticed that a few character variables needed to be converted to factors, such as *Survived*, *Pclass*, and *Sex*. This change would allow the variables to be understood as categories or groups rather than plain text.

I also looked for missing values, with *Age* having 177 missing values and *Cabin* having most of the data missing. For the *Age* variable, the next step was to find its distribution and identify potential outliers with a boxplot shown in the Appendix as Figure 1. Given this, the best approach was to replace the 177 missing values with the median, as median imputation is often preferred over the mean when the data contains significant outliers or is skewed (APXML, 2026). With this new dataset, I created a correlation matrix of the numeric variables to visually see which variables are related to each other (Figure 2). From this, we can see that passengers who had a sibling on board are somewhat likely to have a parent or a child on board; however, this correlation is not that strong.

To continue exploring the data for categorical variables, I created bar charts that showed key information. For instance, Figure 3 shows the sex distribution across the three different passenger classes, showing that the majority of passengers were third class, and most of the third

class was composed of men. Whilst first and second class had a relatively even distribution between men and women. Similarly, looking at Figure 4 and Figure 5, we can see that the proportion of women who survived is much bigger than those who did not, while the proportion of men who died is greater than those who survived. Similarly, we can see that the proportion of people who did not survive is greater in the third class, while the proportion of people who survived is greater in the first class. Finally, we can see in Figure 6 that children between 0 and 12 years had a higher survival proportion than any other age group.

Discussion

The EDA suggested that sex, age, and passenger class might be related to survival. However, to understand how survival outcome depends on other variables using a logistic regression model with backward selection; variables that did not have a significant P-value (<0.05) were dropped until we reached a model with only significant variables: *Passenger Class*, *Sex*, *Age*, *Siblings on Board*. The results of the final model (Model 3) are shown in Table 1. A logistic regression was chosen, as our variable *Survived* is a binary variable. With the final model, we compared its accuracy to the baseline accuracy in predicting non-survival, outperforming the baseline accuracy (shown in Table 2). However, it is important to highlight that the model's accuracy was calculated using the same dataset used to fit the model. Thus, the shown accuracy could overestimate how well the model would perform on new data.

I tested the model by giving it two new passengers that were not included in the original dataset: first, a male, 30 years old, without siblings, from 3rd class; the second passenger, a male, 30 years old, without siblings, but from 1st class. The model gave the 3rd class passenger a 9% chance of survival, and the first passenger a 51% chance of survival. By maintaining the rest of the variables the same, we can still see that class was a key factor in determining survival. Similarly, I continued the testing, this time with 1st class passengers, male, 30 years old, but one with 2 siblings on board, and the other with 1 sibling on board. The model predicted a lower survival chance for the gentleman with two siblings aboard (34%), compared to the passenger with one sibling (43%).

Overall, this paper aimed to answer whether there were any key factors influencing who survived the sinking of the Titanic. The results show that class, sex, age, and a passenger's number of siblings on the ship were key factors that determined survival outcomes. A suggestion for future studies could include a split between training data and testing data for the logistic model, which could improve the model's accuracy when testing new data. Similarly, some factors are not accounted for in the data that could be of importance, such as the mentioned shortage of emergency equipment, which hopefully future studies can include.

Appendix

Figure 1. Boxplot of Age

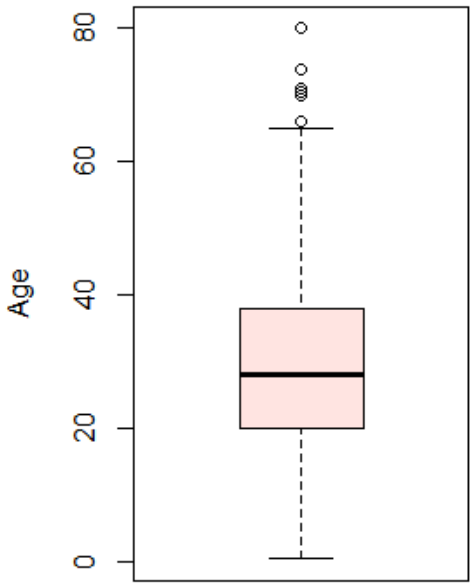
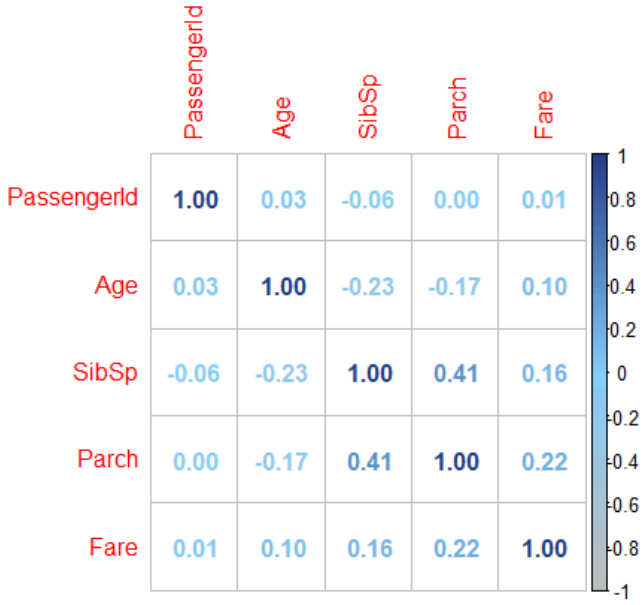
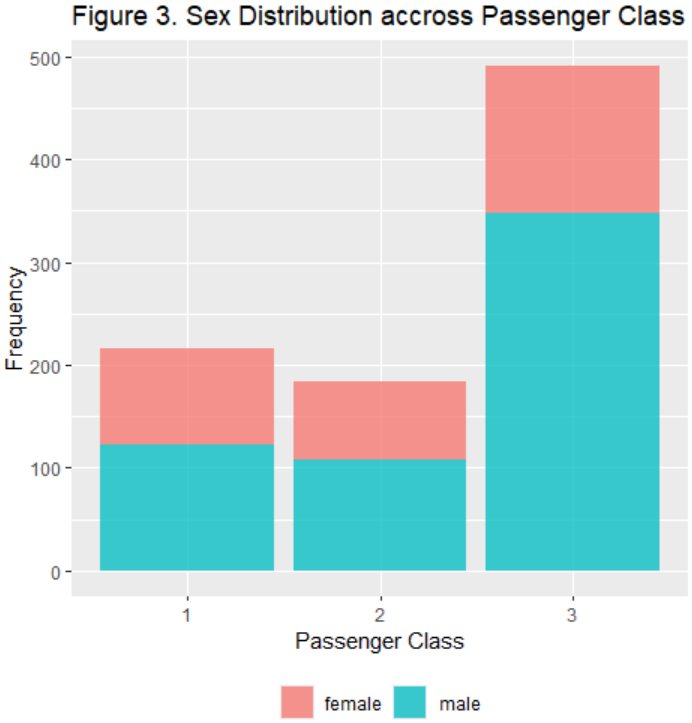
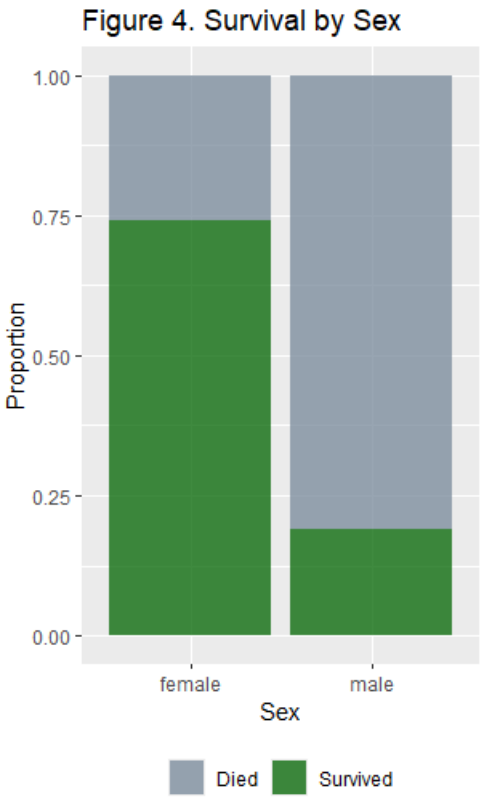
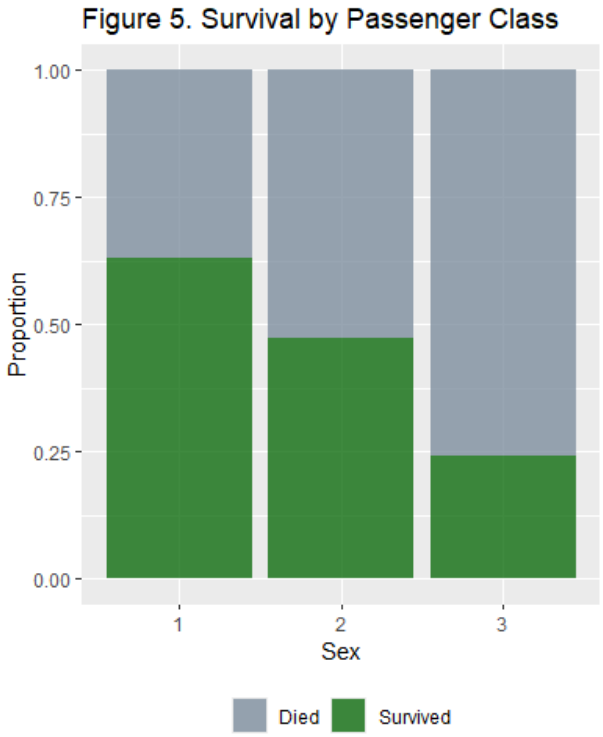


Figure 2. Correlation of Numeric Variables




Figure 3. Sex Distribution accross Passenger Class
Frequency
500
400
300
200
100
0
1
2
3
Passenger Class
female
male


Figure 4. Survival by Sex
Proportion
1.00
0.75
0.50
0.25
0.00
female
male
Sex
Died
Survived


Figure 5. Survival by Passenger Class
Proportion
1.00
0.75
0.50
0.25
0.00
1
2
3
Sex
Died
Survived

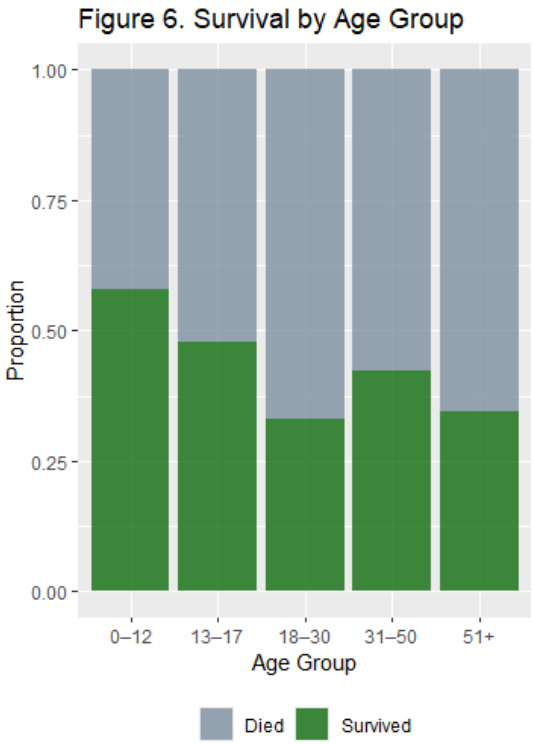

Figure 6. Survival by Age Group
Proportion
1.00
0.75
0.50
0.25
0.00
0–12
13–17
18–30
31–50
51+
Age Group
Died
Survived

Table 1. Model 3 - Estimated Coefficients, Standard Errors, and P-values			
Predictor	Estimated Coefficient	Standard error	P-Value
(Intercept)	4.003	0.397	2 e-16
Passenger 2nd Class	-1.180	0.261	6.28 e-06
Passenger 3rd Class	-2.352	0.243	2 e-16
Male	-2.739	0.194	2 e-16
Age	-0.039	0.007	3.86 e-07
With Siblings	-0.354	0.103	0.0006
AIC	803.23		

Table 2. Baseline Accuracy and Model 3 Accuracy	
Baseline	0.616
Model 3	0.801

References

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